Inverse Planning Demo

This notebook demonstrates the working `memo`-backed inverse-planning pipeline on the default `8x8` gridworld with `3` candidate goals. It covers four concrete steps: inspect the grid, run a few hand-picked scenarios, generate datasets with multiple settings, and train the RNN baselines.

In [10]:
```
import numpy as np
import matplotlib.pyplot as plt
import torch
from dataclasses import replace
from pathlib import Path
from IPython.display import HTML, SVG, display

from inverse_planning.data import collect_dataset, encode_grid_frame, save_c
from inverse_planning.inference import (
    exact_goal_posterior,
    online_goal_posteriors,
    online_posteriors_from_goal_conditioned_action_probs,
    posterior_from_goal_conditioned_scores,
    score_goal_conditioned_policy,
)
from inverse_planning.memo_backend import MemoPolicyBackend
from inverse_planning.rnn_models import ModelFactory
from inverse_planning.simulate import sample_trajectory
from inverse_planning.task import make_default_task
from inverse_planning.training import (
    TrainingConfig,
    build_classifier_loader,
    build_policy_loader,
    evaluate_classifier,
    evaluate_policy_model,
    train_classifier,
    train_policy_model,
)
from inverse_planning.visualize import render_gridworld_svg, render_trajecto

task = make_default_task(beta=2.0)
memo_backend = MemoPolicyBackend(task)
memo_backend.build()
device = 'cuda:0' if torch.cuda.is_available() else 'cpu'
artifacts_dir = Path('artifacts')
artifacts_dir.mkdir(parents=True, exist_ok=True)

print('task shape:', task.shape)
print('goals:', task.goal_locs)
print('training device:', device)

MODEL_STYLES = {
    'action_gru': {'label': 'Action-GRU', 'color': '#1b9e77', 'marker': 'o'}
    'conv_gru': {'label': 'Conv-GRU', 'color': '#d95f02', 'marker': '^'},
```

```
    'goal_conditioned_policy': {'label': 'Goal-Conditioned Policy', 'color':
}

BETA_STYLES = {
    0.75: {'label': 'beta = 0.75', 'color': '#66a61e', 'marker': 'o'},
    1.5: {'label': 'beta = 1.5', 'color': '#e6ab02', 'marker': '^'},
    3.0: {'label': 'beta = 3.0', 'color': '#a6761d', 'marker': 's'},
}

plt.rcParams.update({
    'axes.spines.top': False,
    'axes.spines.right': False,
    'axes.titlesize': 12,
    'axes.labelsize': 11,
    'legend.frameon': False,
})
```

```
task shape: (8, 8)
goals: ((0, 0), (0, 7), (7, 7))
training device: cuda:0
```

1. Grid And Baseline Trajectory

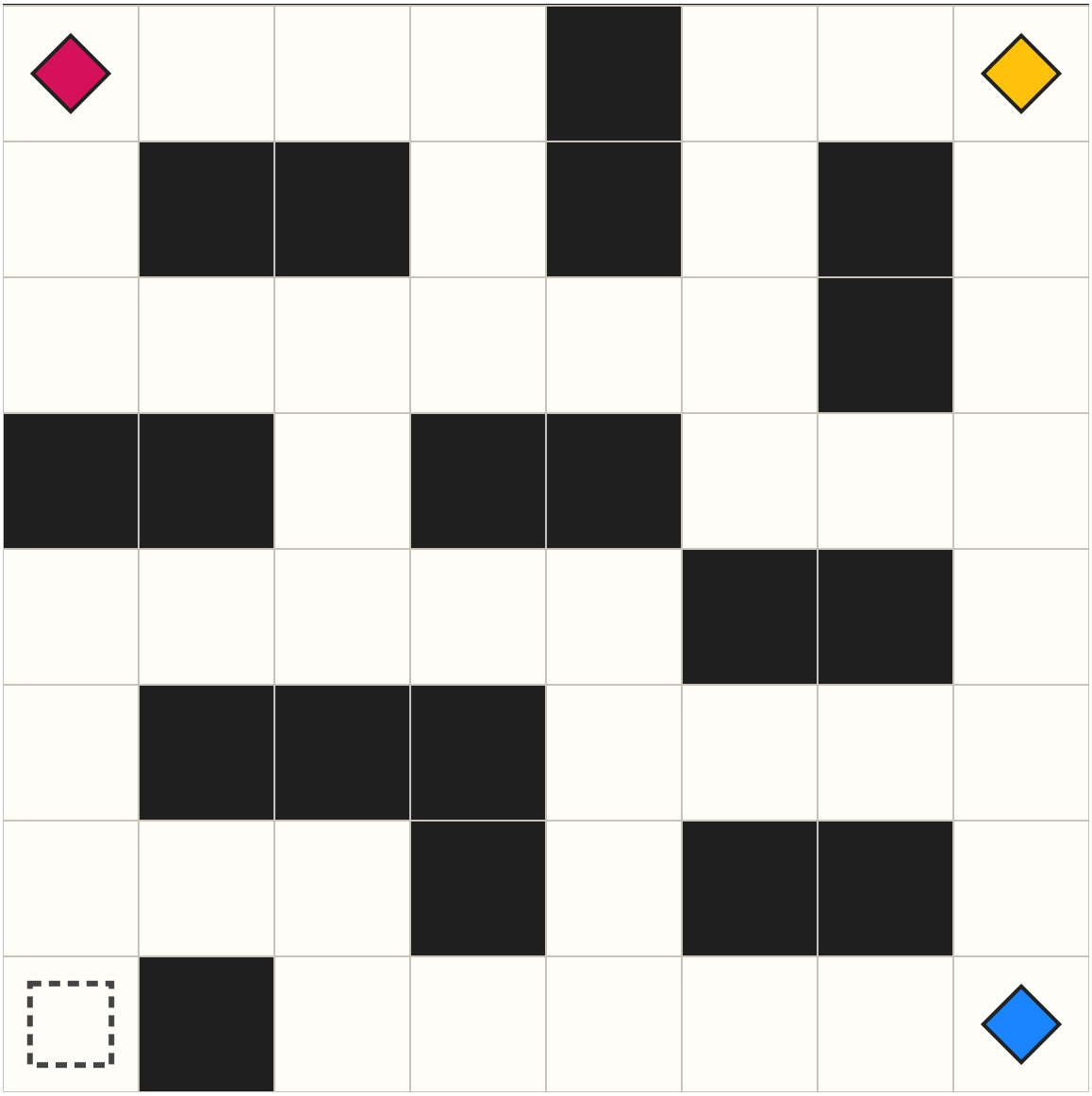
Start by visualizing the default gridworld and one sampled memo-planned trajectory. The exact observer posterior is computed from the action sequence.

In [11]:

```
print('initial location:', task.init_loc)
print(task.grid.astype(int))
display(SVG(render_gridworld_svg(task, cell_size=72, title='Default 8x8 gric

baseline = sample_trajectory(task, horizon=12, rng=np.random.default_rng(0),
baseline_posterior = exact_goal_posterior(task, baseline.action_indices, pol
baseline_online = online_goal_posteriors(task, baseline.action_indices, poli

print('sampled goal index:', baseline.goal_index)
print('sampled actions:', baseline.action_indices)
print('final exact posterior:', np.round(baseline_posterior, 3))
display(SVG(render_gridworld_svg(task, trajectory=baseline, cell_size=72, ti
print('online exact posterior:')
print(np.round(baseline_online, 3))
```

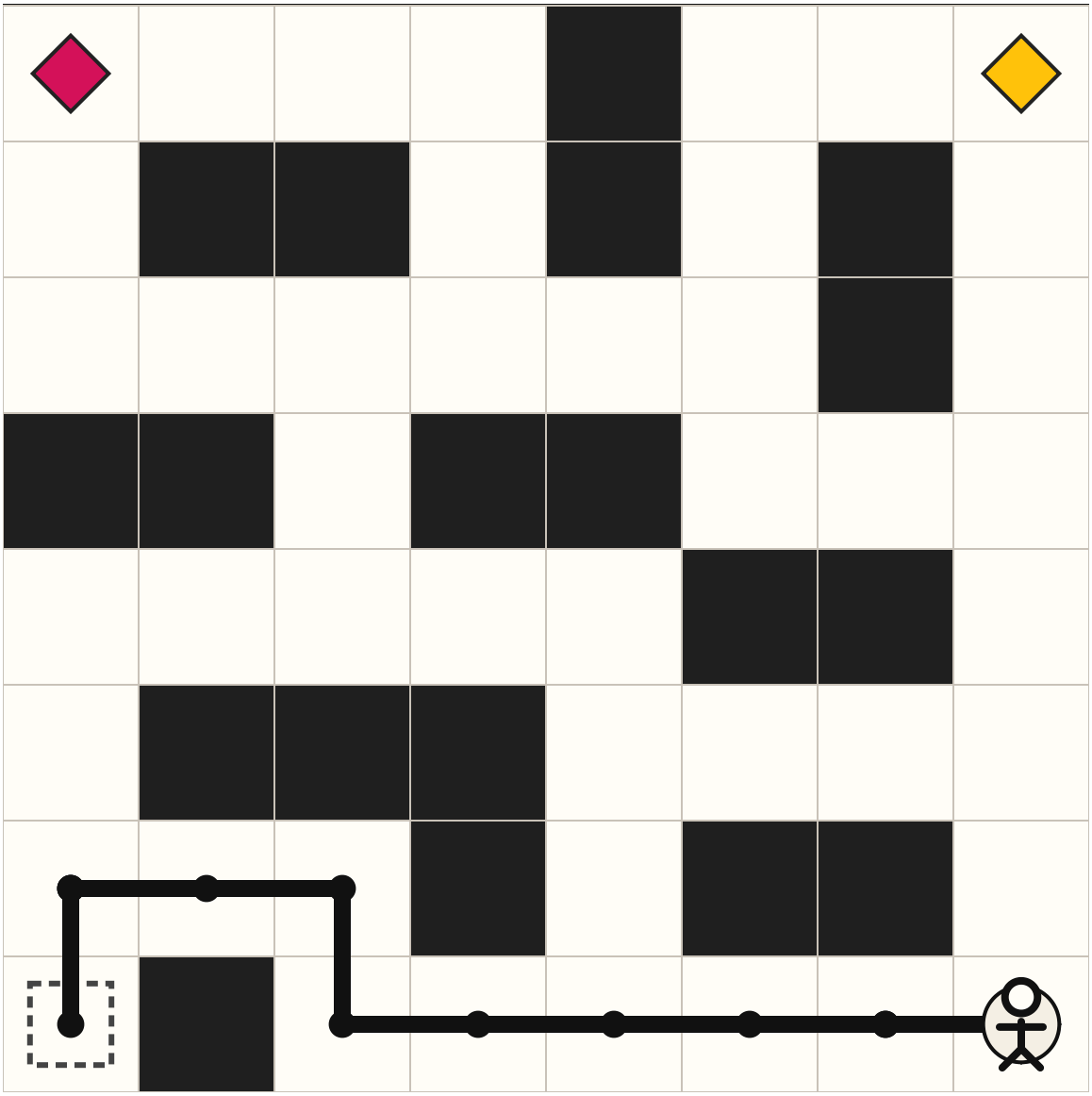
```
initial location: (7, 0)
[[0 0 0 0 1 0 0 0]
 [0 1 1 0 1 0 1 0]
 [0 0 0 0 0 0 1 0]
 [1 1 0 1 1 0 0 0]
 [0 0 0 0 0 1 1 0]
 [0 1 1 1 0 0 0 0]
 [0 0 0 1 0 1 1 0]
 [0 1 0 0 0 0 0 0]]
```

Default 8x8 gridworld



```
sampled goal index: 2
sampled actions: [1, 0, 0, 4, 4, 2, 4, 4, 4, 4, 0, 4]
final exact posterior: [0.    0.005 0.995]
```

Memo-planned baseline trajectory



```
online exact posterior:
[[0.333 0.333 0.333]
 [0.333 0.333 0.333]
 [0.333 0.333 0.333]
 [0.018 0.018 0.965]
 [0.    0.01  0.99 ]
 [0.    0.01  0.99 ]
 [0.    0.01  0.99 ]
 [0.    0.01  0.99 ]
 [0.    0.005 0.995]
 [0.    0.005 0.995]
 [0.    0.005 0.995]
 [0.    0.005 0.995]]
```

2. Scenario Walkthroughs

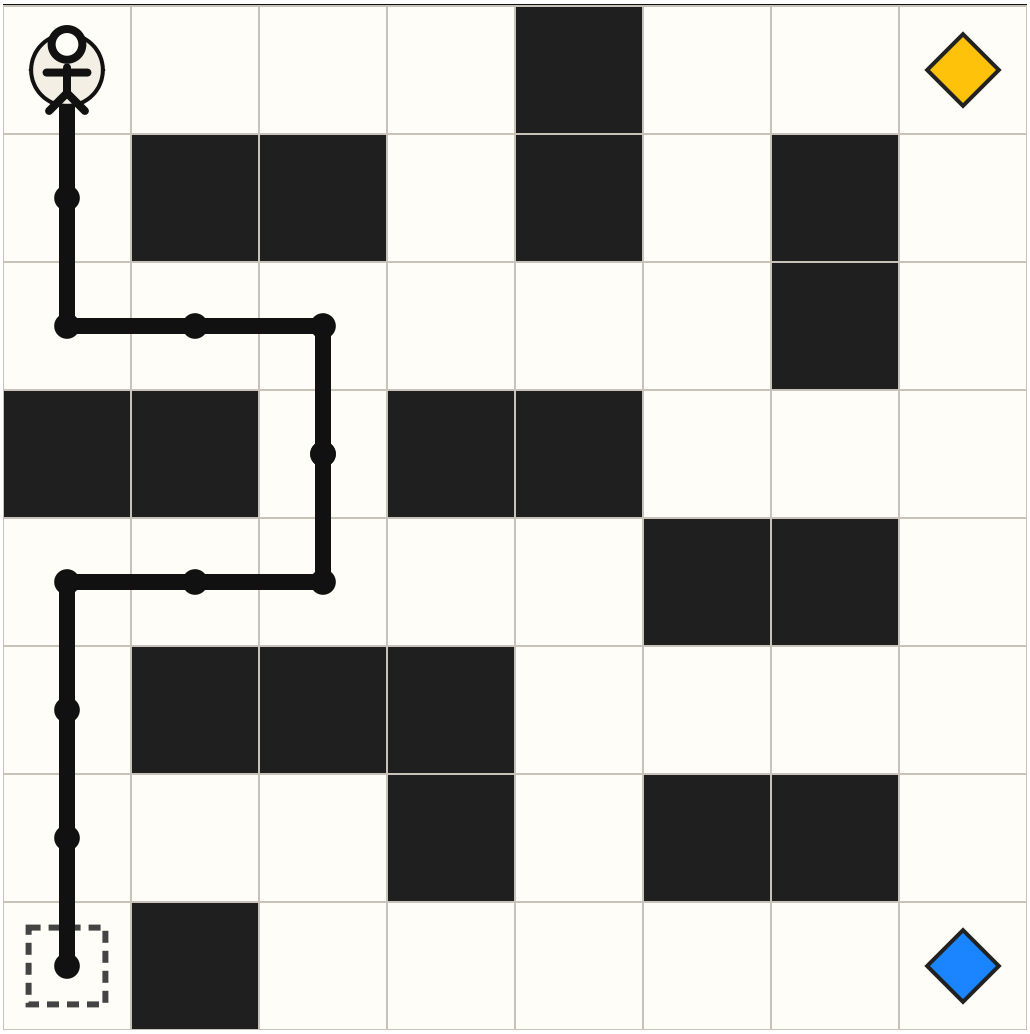
Demonstrate inverse planning on several concrete scenarios. Each scenario changes the start location and fixes the intended goal so the qualitative

posterior updates are easy to inspect.

In [12]:
```
scenario_specs = [
    {'name': 'Northwest chase', 'init_loc': (7, 0), 'goal_index': 0, 'horizo
    {'name': 'Cross-map east goal', 'init_loc': (4, 0), 'goal_index': 1, 'ho
    {'name': 'Bottom-right commitment', 'init_loc': (6, 2), 'goal_index': 2,
]

scenario_runs = []
for spec in scenario_specs:
    scenario_task = replace(task, init_loc=spec['init_loc'])
    scenario_task.validate()
    trajectory = sample_trajectory(
        scenario_task,
        horizon=spec['horizon'],
        rng=np.random.default_rng(spec['seed']),
        goal_index=spec['goal_index'],
        policy_backend=memo_backend,
    )
    posterior = exact_goal_posterior(scenario_task, trajectory.action_indice
    online = online_goal_posteriors(scenario_task, trajectory.action_indices
    scenario_runs.append((spec, scenario_task, trajectory, posterior, online
    print(f"{spec['name']}: start={spec['init_loc']} goal={scenario_task.goa
    print('  actions:', trajectory.action_indices)
    print('  final posterior:', np.round(posterior, 3))
    display(SVG(render_gridworld_svg(scenario_task, trajectory=trajectory, c
    print('  online posterior:')
    print(np.round(online, 3))
    print('-' * 72)
```

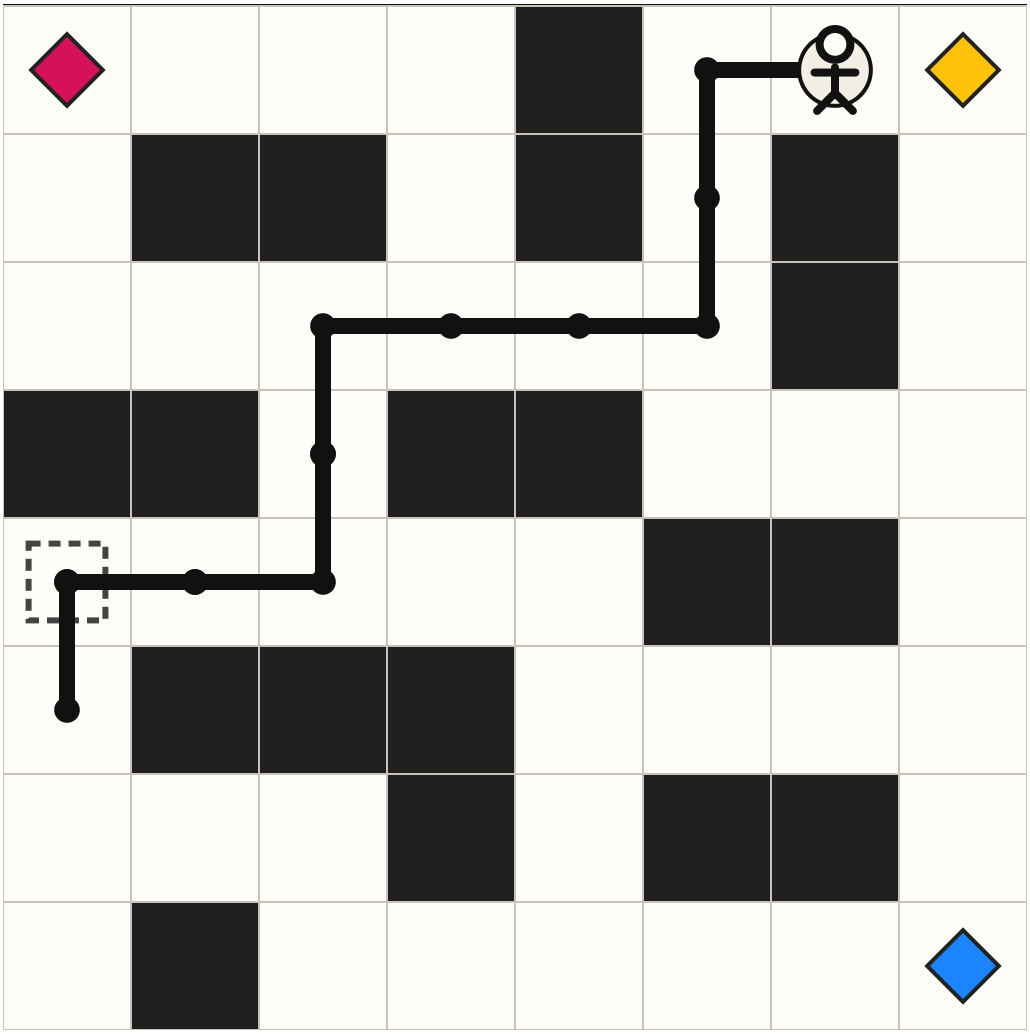
```
Northwest chase: start=(7, 0) goal=(0, 0)
  actions: [1, 1, 1, 4, 4, 1, 0, 1, 3, 3, 1, 1]
  final posterior: [1. 0. 0.]
```

Northwest chase

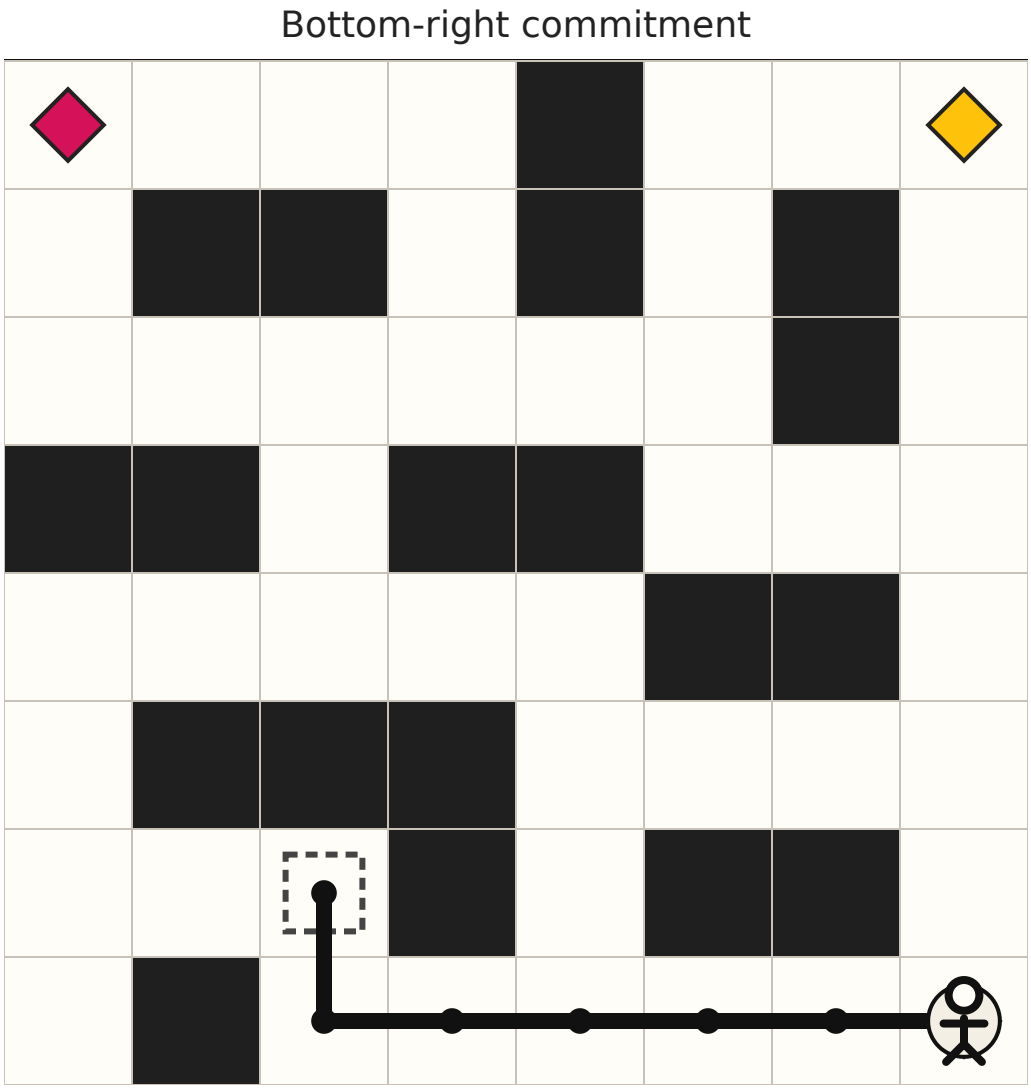


```
  online posterior:
[[0.333 0.333 0.333]
 [0.495 0.495 0.009]
 [0.5   0.5   0.   ]
 [0.5   0.5   0.   ]
 [0.5   0.5   0.   ]
 [0.5   0.5   0.   ]
 [0.5   0.5   0.   ]
 [0.5   0.5   0.   ]
 [0.982 0.018 0.   ]
 [1.    0.    0.   ]
 [1.    0.    0.   ]
 [1.    0.    0.   ]]
--------------------------------------------------------------------------
Cross-map east goal: start=(4, 0) goal=(0, 7)
  actions: [2, 1, 4, 0, 4, 1, 0, 1, 4, 4, 4, 1, 1, 4]
  final posterior: [0. 1. 0.]
```

Cross-map east goal



```
  online posterior:
[[0.032 0.032 0.937]
 [0.394 0.394 0.213]
 [0.436 0.436 0.127]
 [0.436 0.436 0.127]
 [0.436 0.436 0.127]
 [0.499 0.499 0.003]
 [0.499 0.499 0.003]
 [0.5   0.5   0.   ]
 [0.018 0.982 0.   ]
 [0.    1.    0.   ]
 [0.    1.    0.   ]
 [0.    1.    0.   ]
 [0.    1.    0.   ]
 [0.    1.    0.   ]]
--------------------------------------------------------------------------
Bottom-right commitment: start=(6, 2) goal=(7, 7)
  actions: [2, 4, 4, 4, 4, 4, 0, 0, 0, 0]
  final posterior: [0. 0. 1.]
```




```
  online posterior:
[[0.009 0.495 0.495]
 [0.005 0.498 0.498]
 [0.005 0.498 0.498]
 [0.    0.352 0.648]
 [0.    0.352 0.648]
 [0.    0.352 0.648]
 [0.    0.075 0.925]
 [0.    0.012 0.988]
 [0.    0.002 0.998]
 [0.    0.    1.   ]]
--------------------------------------------------------------------------------
```

3. Dataset Budgets

Create several training sets with increasing episode counts, plus one fixed evaluation set. These saved datasets support both the main training run and a clean sample-efficiency study.

In [13]:
```
dataset_specs = [
    {'name': 'budget_64', 'episodes': 64, 'horizon': 12, 'seed': 1},
```

```
    {'name': 'budget_256', 'episodes': 256, 'horizon': 12, 'seed': 2},
    {'name': 'budget_1024', 'episodes': 1024, 'horizon': 12, 'seed': 3},
    {'name': 'budget_2048', 'episodes': 2048, 'horizon': 12, 'seed': 4},
    {'name': 'budget_4096', 'episodes': 4096, 'horizon': 12, 'seed': 5},
    {'name': 'eval_1024', 'episodes': 1024, 'horizon': 12, 'seed': 99},
]

dataset_registry = {}
for spec in dataset_specs:
    bundle = collect_dataset(
        task,
        n_episodes=spec['episodes'],
        horizon=spec['horizon'],
        seed=spec['seed'],
        policy_backend=memo_backend,
    )
    path = artifacts_dir / f"{spec['name']}.npz"
    save_dataset(path, bundle)
    dataset_registry[spec['name']] = {'spec': spec, 'path': path, 'bundle':
    print(spec['name'])
    print('  saved to:', path)
    print('  grids:', bundle.grids.shape)
    print('  actions:', bundle.actions.shape)
    print('  goals:', bundle.goals.shape)
    unique_goals, counts = np.unique(bundle.goals, return_counts=True)
    print('  goal counts:', dict(zip(unique_goals.tolist(), counts.tolist()))
    print()

train_name = 'budget_4096'
eval_name = 'eval_1024'
train_npz = np.load(dataset_registry[train_name]['path'])
eval_npz = np.load(dataset_registry[eval_name]['path'])
print('selected training dataset:', train_name)
print('held-out evaluation dataset:', eval_name)
```

```
budget_64
  saved to: artifacts/budget_64.npz
  grids: (64, 12, 6, 8, 8)
  actions: (64, 12)
  goals: (64,)
  goal counts: {0: 22, 1: 22, 2: 20}

budget_256
  saved to: artifacts/budget_256.npz
  grids: (256, 12, 6, 8, 8)
  actions: (256, 12)
  goals: (256,)
  goal counts: {0: 91, 1: 73, 2: 92}

budget_1024
  saved to: artifacts/budget_1024.npz
  grids: (1024, 12, 6, 8, 8)
  actions: (1024, 12)
  goals: (1024,)
  goal counts: {0: 348, 1: 351, 2: 325}

budget_2048
  saved to: artifacts/budget_2048.npz
  grids: (2048, 12, 6, 8, 8)
  actions: (2048, 12)
  goals: (2048,)
  goal counts: {0: 678, 1: 712, 2: 658}

budget_4096
  saved to: artifacts/budget_4096.npz
  grids: (4096, 12, 6, 8, 8)
  actions: (4096, 12)
  goals: (4096,)
  goal counts: {0: 1422, 1: 1355, 2: 1319}

eval_1024
  saved to: artifacts/eval_1024.npz
  grids: (1024, 12, 6, 8, 8)
  actions: (1024, 12)
  goals: (1024,)
  goal counts: {0: 328, 1: 349, 2: 347}

selected training dataset: budget_4096
held-out evaluation dataset: eval_1024
```

4. Main Training Run

Train all three models on the largest default budget and evaluate them on the fixed held-out split. This gives one strong reference point before the notebook branches into targeted experiments.

In [14]:
```
torch.manual_seed(0)

factory = ModelFactory(
```

```
    in_channels=train_npz['grids'].shape[2],
    n_actions=int(train_npz['actions'].max()) + 1,
    n_goals=train_npz['final_posteriors'].shape[-1],
)
config = TrainingConfig(batch_size=64, epochs=12, learning_rate=5e-4, device

training_runs = {}
for variant in ['action_gru', 'conv_gru']:
    model = factory.build_classifier(variant)
    losses = train_classifier(model, build_classifier_loader(train_npz, batc
    metrics = evaluate_classifier(model, eval_npz, device=device)
    training_runs[variant] = {'model': model, 'losses': losses, 'metrics': m
    torch.save(model.state_dict(), artifacts_dir / f'{variant}.pt')

policy_model = factory.build_policy_model()
policy_losses = train_policy_model(policy_model, build_policy_loader(train_r
policy_metrics = evaluate_policy_model(policy_model, eval_npz, device=device
training_runs['goal_conditioned_policy'] = {'model': policy_model, 'losses':
torch.save(policy_model.state_dict(), artifacts_dir / 'goal_conditioned_poli

for name, run in training_runs.items():
    print(name)
    print('  final loss:', round(run['losses'][-1], 4))
    print('  optimization steps:', len(run['losses']))
    print('  eval final-goal accuracy:', round(run['metrics']['final_goal_ac
    print('  eval posterior KL:', round(run['metrics']['posterior_kl'], 4))
```

```
action_gru
  final loss: 1.1125
  optimization steps: 768
  eval final-goal accuracy: 0.9678
  eval posterior KL: 0.0916
conv_gru
  final loss: 1.2612
  optimization steps: 768
  eval final-goal accuracy: 0.9609
  eval posterior KL: 0.0953
goal_conditioned_policy
  final loss: 0.5262
  optimization steps: 768
  eval final-goal accuracy: 0.9824
  eval posterior KL: 0.0318
```

5. Inference Over Time By Data Budget

Measure how well each model matches the exact observer as more actions are observed. Each curve is an evaluation-set average of posterior KL at timestep `t`, so lower curves indicate faster and more accurate goal inference over time.

In [15]:
```
scaling_budgets = ['budget_64', 'budget_256', 'budget_1024', 'budget_2048',
scaling_config = TrainingConfig(batch_size=64, epochs=5, learning_rate=5e-4,
scaling_results = []

for budget_name in scaling_budgets:
```

```
    budget_npz = np.load(dataset_registry[budget_name]['path'])
    budget_factory = ModelFactory(
        in_channels=budget_npz['grids'].shape[2],
        n_actions=int(budget_npz['actions'].max()) + 1,
        n_goals=budget_npz['final_posteriors'].shape[-1],
    )
    for variant in ['action_gru', 'conv_gru', 'goal_conditioned_policy']:
        torch.manual_seed(0)
        if variant == 'goal_conditioned_policy':
            model = budget_factory.build_policy_model()
            losses = train_policy_model(model, build_policy_loader(budget_np
            metrics = evaluate_policy_model(model, eval_npz, device=device)
        else:
            model = budget_factory.build_classifier(variant)
            losses = train_classifier(model, build_classifier_loader(budget_
            metrics = evaluate_classifier(model, eval_npz, device=device)
        scaling_results.append({
            'budget_name': budget_name,
            'episodes': int(dataset_registry[budget_name]['spec']['episodes'
            'variant': variant,
            'final_loss': float(losses[-1]),
            'posterior_kl': metrics['posterior_kl'],
            'posterior_kl_by_step': metrics['posterior_kl_by_step'],
        })

for row in scaling_results:
    print(f"{row['budget_name']:>11} | {row['variant']:<24} | mean KL = {row
```

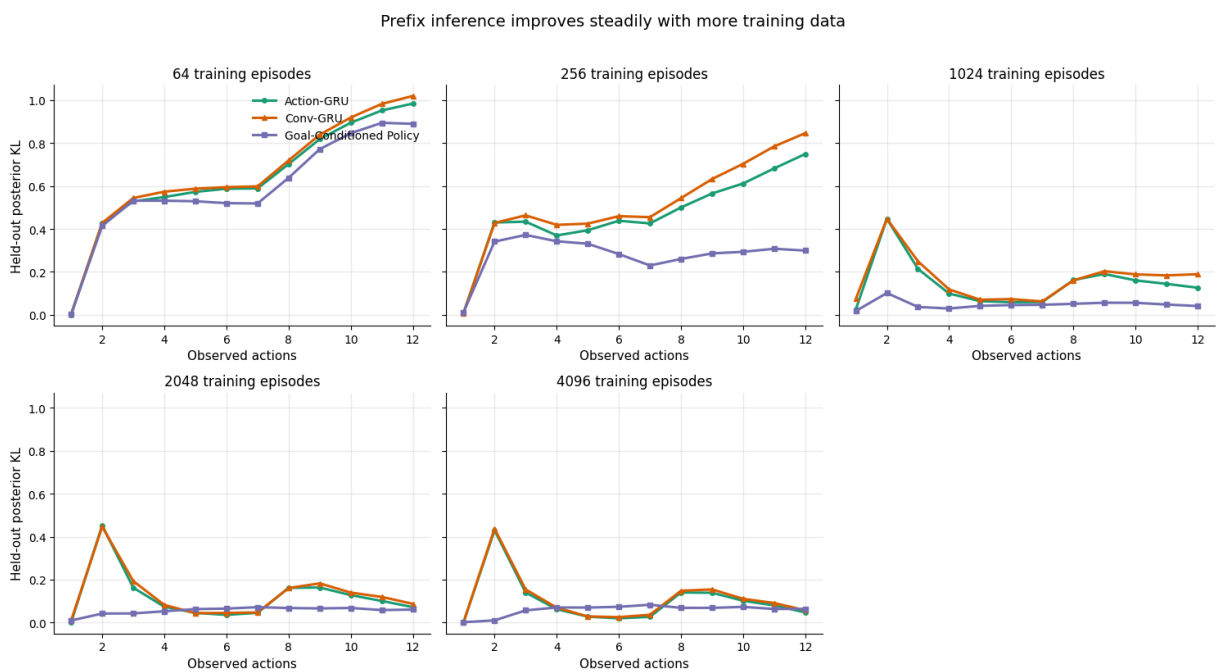
```
  budget_64 | action_gru               | mean KL = 0.6340
  budget_64 | conv_gru                 | mean KL = 0.6510
  budget_64 | goal_conditioned_policy  | mean KL = 0.5908
 budget_256 | action_gru               | mean KL = 0.4675
 budget_256 | conv_gru                 | mean KL = 0.5141
 budget_256 | goal_conditioned_policy  | mean KL = 0.2800
budget_1024 | action_gru               | mean KL = 0.1460
budget_1024 | conv_gru                 | mean KL = 0.1685
budget_1024 | goal_conditioned_policy  | mean KL = 0.0478
budget_2048 | action_gru               | mean KL = 0.1201
budget_2048 | conv_gru                 | mean KL = 0.1298
budget_2048 | goal_conditioned_policy  | mean KL = 0.0556
budget_4096 | action_gru               | mean KL = 0.1012
budget_4096 | conv_gru                 | mean KL = 0.1095
budget_4096 | goal_conditioned_policy  | mean KL = 0.0586
```

6. Goal Inference Over Time

Each panel corresponds to one training-data budget. The y-axis is held-out posterior KL, averaged over episodes at each timestep. Lower values mean the model stays closer to the exact inverse-planning posterior as evidence accumulates over time.

In [16]:
```
fig, axes = plt.subplots(2, 3, figsize=(15, 8), sharey=True)
axes = axes.flatten()
for ax, budget_name in zip(axes, scaling_budgets):
```

```
    rows = [row for row in scaling_results if row['budget_name'] == budget_r
    rows = sorted(rows, key=lambda row: row['episodes'])
    for row in rows:
        style = MODEL_STYLES[row['variant']]
        timesteps = np.arange(1, len(row['posterior_kl_by_step']) + 1)
        ax.plot(
            timesteps,
            row['posterior_kl_by_step'],
            marker=style['marker'],
            markersize=4,
            linewidth=2.2,
            color=style['color'],
            label=style['label'],
        )
    n_eps = dataset_registry[budget_name]['spec']['episodes']
    ax.set_title(f'{n_eps} training episodes')
    ax.set_xlabel('Observed actions')
    ax.grid(alpha=0.25)

axes[0].set_ylabel('Held-out posterior KL')
axes[3].set_ylabel('Held-out posterior KL')
axes[0].legend(loc='upper right', ncol=1)
if len(axes) > len(scaling_budgets):
    for ax in axes[len(scaling_budgets):]:
        ax.axis('off')
fig.suptitle('Goal inference improves steadily with more training data', y=1
fig.tight_layout()
plt.show()
```



7. Rationality Ablation

Vary the planner rationality parameter `beta` while keeping the grid and candidate goals fixed. Lower `beta` produces noisier behavior, which should

make inverse planning harder.

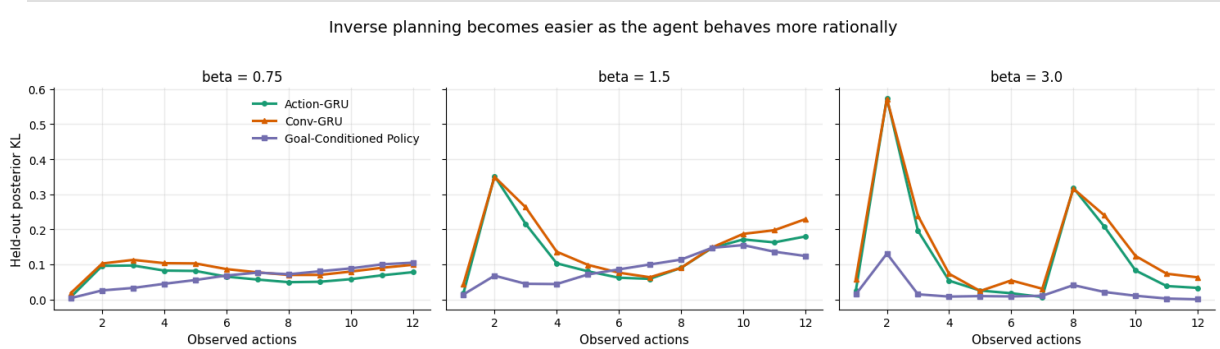
In [17]:
```
beta_values = [0.75, 1.5, 3.0]
beta_train_episodes = 1024
beta_eval_episodes = 512
beta_results = []

for beta in beta_values:
    beta_task = make_default_task(beta=beta)
    beta_backend = MemoPolicyBackend(beta_task)
    beta_backend.build()
    beta_train = collect_dataset(beta_task, n_episodes=beta_train_episodes,
    beta_eval = collect_dataset(beta_task, n_episodes=beta_eval_episodes, ho

    beta_train_path = artifacts_dir / f'beta_{str(beta).replace(".", "p")}_t
    beta_eval_path = artifacts_dir / f'beta_{str(beta).replace(".", "p")}_ev
    save_dataset(beta_train_path, beta_train)
    save_dataset(beta_eval_path, beta_eval)
    beta_train_npz = np.load(beta_train_path)
    beta_eval_npz = np.load(beta_eval_path)
    beta_factory = ModelFactory(
        in_channels=beta_train_npz['grids'].shape[2],
        n_actions=int(beta_train_npz['actions'].max()) + 1,
        n_goals=beta_train_npz['final_posteriors'].shape[-1],
    )
    beta_config = TrainingConfig(batch_size=64, epochs=5, learning_rate=5e-4

    for variant in ['action_gru', 'conv_gru', 'goal_conditioned_policy']:
        torch.manual_seed(0)
        if variant == 'goal_conditioned_policy':
            model = beta_factory.build_policy_model()
            losses = train_policy_model(model, build_policy_loader(beta_trai
            metrics = evaluate_policy_model(model, beta_eval_npz, device=dev
        else:
            model = beta_factory.build_classifier(variant)
            losses = train_classifier(model, build_classifier_loader(beta_tr
            metrics = evaluate_classifier(model, beta_eval_npz, device=devic
        beta_results.append({
            'beta': beta,
            'variant': variant,
            'posterior_kl': metrics['posterior_kl'],
            'posterior_kl_by_step': metrics['posterior_kl_by_step'],
            'final_loss': float(losses[-1]),
        })

for row in beta_results:
    print(f"beta={row['beta']:<4} | {row['variant']:<24} | mean KL = {row['p
```

```
beta=0.75 | action_gru               | mean KL = 0.0658
beta=0.75 | conv_gru                 | mean KL = 0.0844
beta=0.75 | goal_conditioned_policy  | mean KL = 0.0627
beta=1.5  | action_gru               | mean KL = 0.1366
beta=1.5  | conv_gru                 | mean KL = 0.1569
beta=1.5  | goal_conditioned_policy  | mean KL = 0.0919
beta=3.0  | action_gru               | mean KL = 0.1314
beta=3.0  | conv_gru                 | mean KL = 0.1556
beta=3.0  | goal_conditioned_policy  | mean KL = 0.0226
```

8. Rationality Curves

These curves show how posterior quality changes when the acting agent is less or more rational. Lower `beta` means noisier action choices, so the inference problem should stay ambiguous for longer prefixes.

In [18]:
```
fig, axes = plt.subplots(1, 3, figsize=(15, 4), sharey=True)
for ax, beta in zip(axes, beta_values):
    rows = [row for row in beta_results if row['beta'] == beta]
    for row in rows:
        style = MODEL_STYLES[row['variant']]
        timesteps = np.arange(1, len(row['posterior_kl_by_step']) + 1)
        ax.plot(
            timesteps,
            row['posterior_kl_by_step'],
            marker=style['marker'],
            markersize=4,
            linewidth=2.2,
            color=style['color'],
            label=style['label'],
        )
    ax.set_title(BETA_STYLES[beta]['label'])
    ax.set_xlabel('Observed actions')
    ax.grid(alpha=0.25)
axes[0].set_ylabel('Held-out posterior KL')
axes[0].legend(loc='upper right')
fig.suptitle('Inverse planning becomes easier as the agent behaves more rati
fig.tight_layout()
plt.show()
```



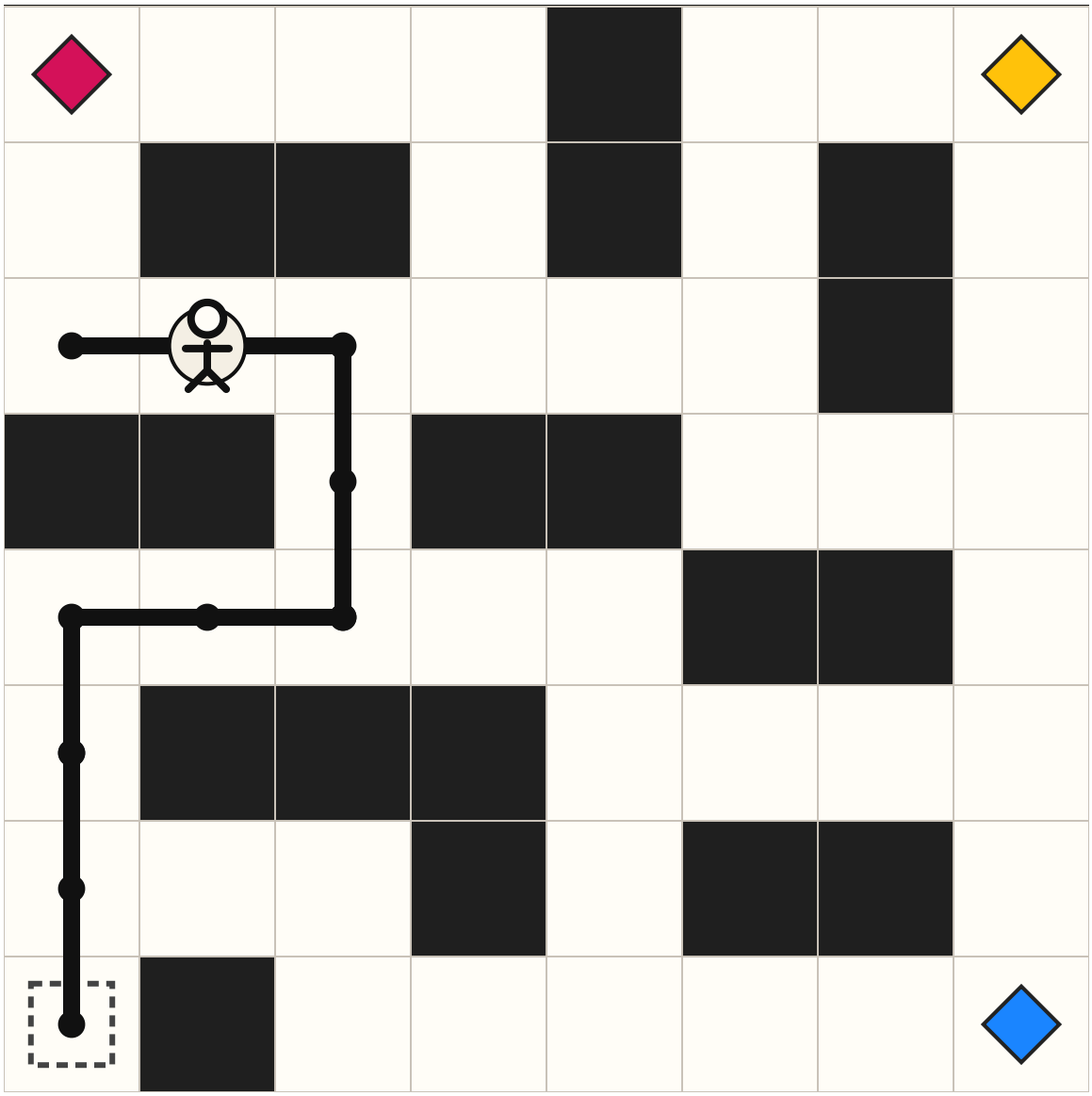
9. Held-Out Trajectory Comparison

Return to one held-out trajectory from the default task and compare the exact posterior path against the models trained on the largest dataset.

In [19]:
```
heldout = sample_trajectory(task, horizon=12, rng=np.random.default_rng(21),
heldout_exact = online_goal_posteriors(task, heldout.action_indices, policy_

frames = np.stack(
    [encode_grid_frame(task, position=heldout.positions[step], step=step, hc
    axis=0,
)[None]
prev_actions = np.full((1, len(heldout.action_indices)), task.n_actions, dty
prev_actions[0, 1:] = np.asarray(heldout.action_indices[:-1], dtype=np.int64
action_indices = np.asarray(heldout.action_indices, dtype=np.int64)

frames_t = torch.tensor(frames, dtype=torch.float32, device=device)
prev_actions_t = torch.tensor(prev_actions, dtype=torch.long, device=device)

for run in training_runs.values():
    run['model'].eval()
    run['model'].to(device)

with torch.no_grad():
    action_gru_out = training_runs['action_gru']['model'](frames_t, prev_act
    conv_gru_out = training_runs['conv_gru']['model'](frames_t, prev_actions

action_gru_final = torch.softmax(action_gru_out['goal_logits'], dim=-1)[0].c
conv_gru_final = torch.softmax(conv_gru_out['goal_logits'], dim=-1)[0].detac
action_gru_online = torch.softmax(action_gru_out['posterior_logits'], dim=-1
conv_gru_online = torch.softmax(conv_gru_out['posterior_logits'], dim=-1)[0]

policy_action_probabilities = []
policy_scores = []
with torch.no_grad():
    for goal_index in range(task.n_goals):
        goal_ids = torch.tensor([goal_index], dtype=torch.long, device=devic
        logits = training_runs['goal_conditioned_policy']['model'](frames_t,
        probs = torch.softmax(logits, dim=-1)[0].detach().cpu().numpy()
        policy_action_probabilities.append(probs)
        policy_scores.append(score_goal_conditioned_policy(probs, action_inc
policy_action_probabilities = np.asarray(policy_action_probabilities)
policy_posterior = posterior_from_goal_conditioned_scores(np.asarray(policy_
policy_online = online_posteriors_from_goal_conditioned_action_probs(policy_

print('held-out goal index:', heldout.goal_index)
print('held-out actions:', heldout.action_indices)
print('exact final posterior:', np.round(heldout_exact[-1], 3))
print('action-gru final belief:', np.round(action_gru_final, 3))
print('conv-gru final belief:', np.round(conv_gru_final, 3))
print('goal-conditioned policy posterior:', np.round(policy_posterior, 3))

display(SVG(render_gridworld_svg(task, trajectory=heldout, cell_size=72, tit
```

```
held-out goal index: 0
held-out actions: [1, 1, 0, 1, 4, 4, 0, 1, 1, 3, 3, 4]
exact final posterior: [0.982 0.018 0.   ]
action-gru final belief: [0.998 0.001 0.   ]
conv-gru final belief: [0.999 0.001 0.   ]
goal-conditioned policy posterior: [0.998 0.002 0.   ]
```

Held-out memo-generated trajectory

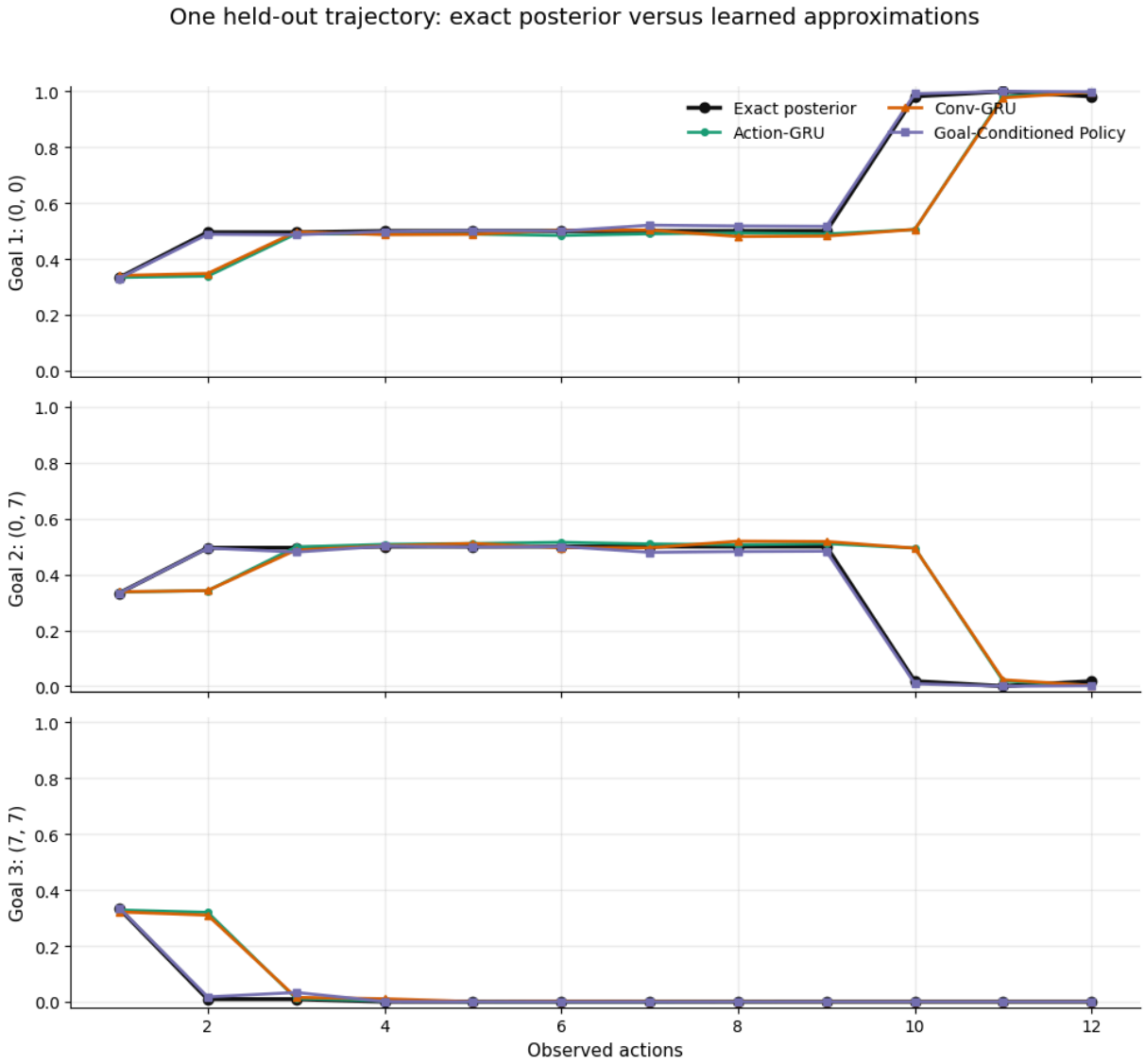


10. Posterior Curves For One Trajectory

The curves below show the exact posterior and learned approximations on a single held-out sequence. Unlike the aggregate prefix plots above, these are useful for qualitative inspection of where the models change their minds.

In [20]:
```
timesteps = np.arange(1, len(heldout.action_indices) + 1)
goal_labels = [f'Goal {idx + 1}: {goal}' for idx, goal in enumerate(task.goa
fig, axes = plt.subplots(task.n_goals, 1, figsize=(10, 3 * task.n_goals), sh
if task.n_goals == 1:
```

```
    axes = [axes]
for goal_index, ax in enumerate(axes):
    ax.plot(timesteps, heldout_exact[:, goal_index], marker='o', linewidth=2
    for variant, series in [
        ('action_gru', action_gru_online[:, goal_index]),
        ('conv_gru', conv_gru_online[:, goal_index]),
        ('goal_conditioned_policy', policy_online[:, goal_index]),
    ]:
        style = MODEL_STYLES[variant]
        ax.plot(
            timesteps,
            series,
            marker=style['marker'],
            markersize=4,
            linewidth=1.9,
            color=style['color'],
            label=style['label'],
        )
    ax.set_ylim(-0.02, 1.02)
    ax.set_ylabel(goal_labels[goal_index])
    ax.grid(alpha=0.25)
    if goal_index == 0:
        ax.legend(loc='upper right', ncol=2)
axes[-1].set_xlabel('Observed actions')
fig.suptitle('One held-out trajectory: exact posterior versus learned approx
fig.tight_layout()
plt.show()
```


One held-out trajectory: exact posterior versus learned approximations
Exact posterior
Action-GRU
Conv-GRU
Goal-Conditioned Policy
Goal 1: (0, 0)
Goal 2: (0, 7)
Goal 3: (7, 7)
Observed actions
0.0
0.2
0.4
0.6
0.8
1.0
2
4
6
8
10
12